

Small world models learn motion before collisions

Random experience, short-horizon prediction, and the limits of ensemble agreement

Abstract

A three-member probabilistic ensemble learns useful motion predictions from 500–2,000 random-action transitions. In two small puck worlds, all 24 fitted models outperform constant-state prediction at one and eight steps; 20 do so at 24 steps. Residual error concentrates on rare contact-like transitions. Longer action holds increase spatial coverage, but coverage and ensemble agreement do not reliably establish accurate long-range dynamics.

Methods

We cross two fixed layouts ([open](#), [rooms](#)), uniform random action holds of 1 or 8 primitive frames, and training sets of 500 or 2,000 actual transitions. Smaller sets are prefixes of longer trajectories. Fitting budgets match; bootstrap batches are independent.

Each ensemble has three two-layer, 64-unit networks. Continuous heads predict Gaussian state changes; a Bernoulli head predicts the door flag. Checkpoints precede fitting and follow 100 and 500 optimizer batches. Evaluation uses 32 separate actual trajectories of 48 frames, from 16 initial positions, with both action-hold policies. Probe experience never enters replay.

SCOPE OF EVIDENCE

24 fits, three seeds per condition. This is a development diagnostic, with no confirmatory significance claim.

No goal, reward, pretrained model, imagined training examples, or task-directed collection.

WHAT WAS HELD FIXED

Adam 0.002; batch 128 per member; normalized state/delta scales $[1, 1, \frac{1}{2}, \frac{1}{2}, 1, 1]$; log-variance bounds $[-8, 1]$; binary loss weight 10.

Held-out starts differ from the collection start, but may overlap regions later visited. This is fixed-layout generalization.

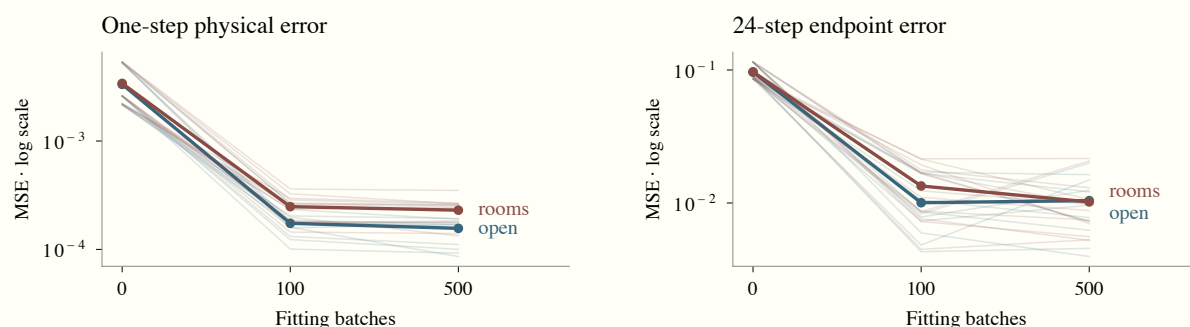


Figure 1. Fitting reduces error sharply before 100 batches; another 400 updates do not uniformly improve long rollouts. Fine lines are individual runs, with means in blue (open) and red (rooms). Physical MSE averages x , y , v_x , v_y in declared observation units. The endpoint at 24 frames is 1.2 seconds ahead.

The missing events matter

Transitions in which a moving velocity coordinate suddenly reaches zero account for around 1% of probes, yet roughly 70% of the remaining physical squared error. Their mean per-event error is about 300–350 times that of other transitions. Small aggregate loss therefore conceals events that can invalidate plans near walls.

Long-rollout position RMSE remains 0.070–0.193 per coordinate in a unit-square world. Predictive improvement is evidence of learned motion; it is not evidence that the world is mastered.

OBSERVED CONTACT PROXY

open

0.74% → 69%

transitions → total squared error

rooms

1.06% → 74%

transitions → total squared error

Velocity magnitude >0.025 becomes $<10^{-7}$ on at least one axis. These are observed transitions, not oracle collision labels.

Coverage is easier to obtain than reliable forecasts

World	Hold frames	Training transitions	Coverage 256 cells	1-step MSE $\times 10^{-4}$	8-step MSE $\times 10^{-3}$	24-step MSE $\times 10^{-3}$
open	1	500	9.8%	1.46	1.28	13.43
open	1	2,000	27.1%	1.43	1.17	9.79
open	8	500	19.8%	1.79	1.73	8.82
open	8	2,000	60.0%	1.58	1.52	9.64
rooms	1	500	9.5%	2.15	1.05	11.24
rooms	1	2,000	24.2%	2.09	0.97	7.70
rooms	8	500	18.0%	2.67	2.52	13.87
rooms	8	2,000	54.8%	2.30	1.55	7.93

Means over three seeds after 500 fitting batches. Spatial coverage includes all cells in the enclosing square. Holding an action repeats actual primitive interactions; it does not accelerate simulated time.

Agreement is an imperfect signal

Local disagreement and real one-step error have Spearman correlations from -0.167 to 0.496; 5 of 24 are negative. The most-disagreed decile is error-enriched in the group means, but the ranking is inconsistent.



Figure 2. One dot per seed. Row labels give world, action hold, and training transitions. Members receive the same actual state/action when disagreement is computed. Lines connect the range across three seeds, not a confidence interval.

A bounded next experiment

Median fitting time is **68 ms for 100 batches** and **335 ms for 500** on one CPU thread. The longer fit improves 24-step error in 16/24 runs and worsens it in the others. Online comparisons should count warmup and fitting, and compare planning horizons without assuming an optimum.

Prioritize real-contact prediction, generic replay sampling of high observed errors, and closed-loop control checks. Keep the evaluation stream separate. Adding hidden exact-wall constraints would no longer test the learned simulator.

Reproduce. `python -m rtga.model_study --output results/002-model-calibration`, then `python results/002-model-calibration/analyze.py`. Use the repository environment.

Complete records include all configurations, per-transition errors, uncertainty values, and runtime provenance. Aggregates, actual trajectories, 24 checkpoints, and exact source snapshots are retained. The longer note states definitions and limitations.

CALIBRATION LIMITS

These are error-ranking checks, not probability calibration or evidence of information gain. A separate experiment must test whether selected transitions produce subsequent learning progress.

NO MECHANISM LEARNED

The door is always open in both layouts; the noise sensor is constant. Tiny Brier scores therefore test a constant flag. This study contains no switch event and cannot establish noisy-TV avoidance.

PROVENANCE

Timing excludes collection and evaluation. Fits were not continued to convergence.

The constant-state baseline and contact breakdown were added descriptively after the initial run. Repeated prediction metrics matched exactly; model settings were unchanged.

SOURCE SHA-256 PREFIXES

models 1d31a88ba68e
envs 9883f95d36f3
study ea3a4f9aee5e

Full hashes appear in the records.